

Laporan Hands-on Docker — Cloud Computing

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Repo : <https://github.com/fhanyuh/cloud-computing-docker-101012300238>

Bagian 1 — Lifecycle Kontainer ubuntu:22.04

Perintah

```
docker run -i --name tes-ubuntu-101012300238 ubuntu:22.04 bash -c 'cat /etc/os-release'
docker ps -a --filter name=tes-ubuntu-101012300238
docker rm tes-ubuntu-101012300238
```

Output

```
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS		
ff5ff4647379	ubuntu:22.04	"bash -c 'cat /etc/o..."	1 second ago	Exited (0) Less		
tes-ubuntu-101012300238						
tes-ubuntu-101012300238						
Status after rm:						
CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES

Bagian 2 — app.py dan Dockerfile

app.py

```
import http.server
import socketserver
import os

PORT = 8000

NIM = os.environ.get("STUDENT_NIM", "UNKNOWN")

class MyHandler(http.server.SimpleHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header("Content-type", "text/html; charset=utf-8")
        self.end_headers()
        html = f"""<!DOCTYPE html>
<html>
<head>
<title>Praktikum Cloud Computing - Docker</title>
```

```

<style>

    body {{ font-family: system-ui, -apple-system, sans-serif; background: #0f172a;
color: #f8fafc; display: flex; justify-content: center; align-items: center; height:
100vh; margin: 0; }}

    .card {{ background: #1e293b; padding: 2rem; border-radius: 12px; box-shadow: 0
10px 25px rgba(0,0,0,0.5); text-align: center; border: 1px solid #334155; }}

    h1 {{ color: #38bdf8; margin-bottom: 0.5rem; }}

    p {{ font-size: 1.25rem; color: #94a3b8; }}

    .nim {{ color: #4ade80; font-weight: bold; font-size: 1.5rem; }}

    .badge {{ display: inline-block; padding: 0.25rem 0.75rem; background: #0284c7;
color: white; border-radius: 9999px; font-size: 0.875rem; margin-top: 1rem; }}

</style>

</head>

<body>

    <div class="card">

        <h1>Praktikum Docker Berhasil!</h1>

        <p>Aplikasi web berjalan di dalam kontainer.</p>

        <p>Dikembangkan oleh NIM: <span class="nim">{NIM}</span></p>

        <div class="badge">Cloud Computing BS1TT-47</div>

    </div>

</body>

</html>""

self.wfile.write(html.encode("utf-8"))

def log_message(self, format, *args):

    # Clean logging for container stdout

    print(f"[{self.log_date_time_string()}] {format % args}")

```

```

if __name__ == "__main__":

    with socketserver.TCPServer(("", PORT), MyHandler) as httpd:

        print(f"Serving HTTP on port {PORT} with STUDENT NIM={NIM}")

        httpd.serve_forever()

```

Dockerfile

```

FROM python:3.11-slim

WORKDIR /app

ENV PYTHONUNBUFFERED=1

ENV STUDENT_NIM=101012300238

COPY app.py /app/app.py

EXPOSE 8000

CMD ["python", "/app/app.py"]

```

Instruksi	Fungsi
FROM python:3.11-slim	Base image Python minimal untuk menjalankan app.py
WORKDIR /app	Menetapkan direktori kerja di dalam kontainer
ENV PYTHONUNBUFFERED=1	Menonaktifkan buffering stdout agar log langsung tampil
ENV STUDENT_NIM=101012300238	Menyuntikkan NIM sebagai environment variable ke aplikasi
COPY app.py /app/app.py	Menyalin source code aplikasi ke image
EXPOSE 8000	Mendeklarasikan port yang digunakan aplikasi di dalam kontainer
CMD ["python", "/app/app.py"]	Perintah default yang dijalankan saat kontainer start

Build

```

docker build -t web-tugas-101012300238:1.0 .

#8 exporting manifest list
sha256:e3087ae519052f3f60b23f7af0ad1b9a5105258e4da624498c741d8241708eee 0.0s done

#8 naming to docker.io/library/web-tugas-101012300238:1.0 done

```

```
#8 unpacking to docker.io/library/web-tugas-101012300238:1.0 done
#8 DONE 0.1s
```

Bagian 3 — Docker Compose (service web dan cache)

docker-compose.yml

```
services:
  web:
    build: .
    image: web-tugas-101012300238:1.0
    container_name: web-container-101012300238
    ports:
      - "8080:8000"
    environment:
      STUDENT_NIM: "101012300238"
    depends_on:
      - cache
    restart: unless-stopped
  cache:
    image: redis:7-alpine
    container_name: cache-container-101012300238
    restart: unless-stopped
```

Menjalankan

```
docker compose up -d
```

Screenshot 1 — docker compose ps

```
docker compose ps
```

Service	Status	Ports
web	Up	0.0.0.0:8080->8000/tcp
cache	Up	6379/tcp

NAME	IMAGE	COMMAND	SERVICE
cache-container-101012300238	redis:7-alpine	"docker-entrypoint.s..."	cache
web-container-101012300238	web-tugas-101012300238:1.0	"python /app/app.py"	web

Screenshot 2 — curl http://localhost:8080

```
curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
  <title>Praktikum Cloud Computing - Docker</title>
  <meta charset="UTF-8">
</head>
<body>
  <div class="card">
    <h1>Praktikum Docker Berhasil!</h1>
    <p>Aplikasi web berjalan di dalam kontainer.</p>
    <p>Dikembangkan oleh NIM: <span class="nim">101012300238</span></p>
    <div class="badge">Cloud Computing BS1TT-47</div>
  </div>
</body>
</html>
```



```
<!DOCTYPE html>
<html>
<head>
  <title>Praktikum Cloud Computing - Docker</title>
  <style>
    body { font-family: system-ui, -apple-system, sans-serif; background: #0f172a; color: white; }
    .card { background: #1e293b; padding: 2rem; border-radius: 12px; box-shadow: 0 10px 15px #000; }
    h1 { color: #38bdf8; margin-bottom: 0.5rem; }
    p { font-size: 1.25rem; color: #94a3b8; }
    .nim { color: #4ade80; font-weight: bold; font-size: 1.5rem; }
    .badge { display: inline-block; padding: 0.25rem 0.75rem; background: #0284c7; color: white; border-radius: 6px; }
  </style>
</head>
<body>
  <div class="card">
    <h1>Praktikum Docker Berhasil!</h1>
    <p>Aplikasi web berjalan di dalam kontainer.</p>
    <p>Dikembangkan oleh NIM: <span class="nim">101012300238</span></p>
    <div class="badge">Cloud Computing BS1TT-47</div>
  </div>
</body>
</html>
```

Menghentikan

```
docker compose down
```